AI Document Assistant

Spring Boot + React Based RAG System

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Technology Stack: Spring Boot, React, MySQL, Docker, GitHub Actions

This project is an AI-powered document assistant that allows users to upload PDF documents, extract document content, store chunks in MySQL, and ask questions based on uploaded content.

1. Features

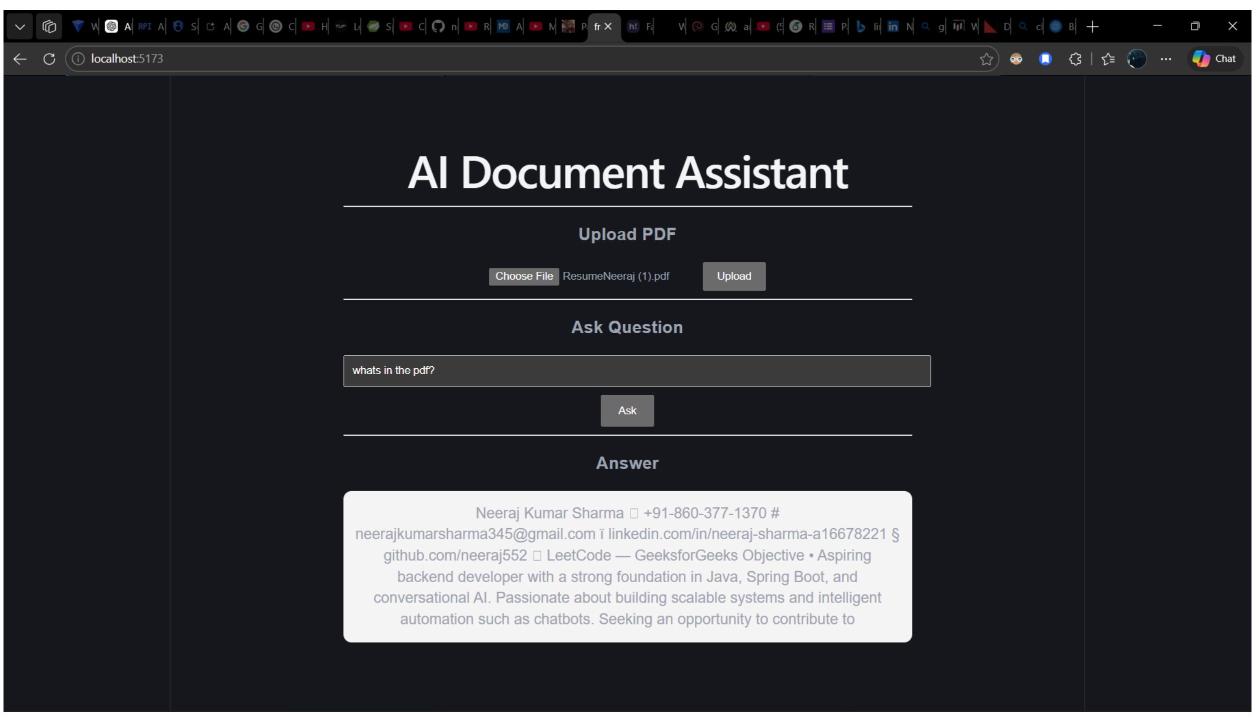
Feature	Description
PDF Upload	Users can upload PDF files through React frontend
Text Extraction	PDF content extracted using Apache PDFBox
Chunking	Large document text divided into smaller chunks
Question Answering	Context-based question answering system
Database Storage	File metadata and chunks stored in MySQL
Frontend	Built using React + Vite
CI/CD	GitHub Actions workflow integration
Docker Ready	Backend prepared for containerization

2. System Architecture

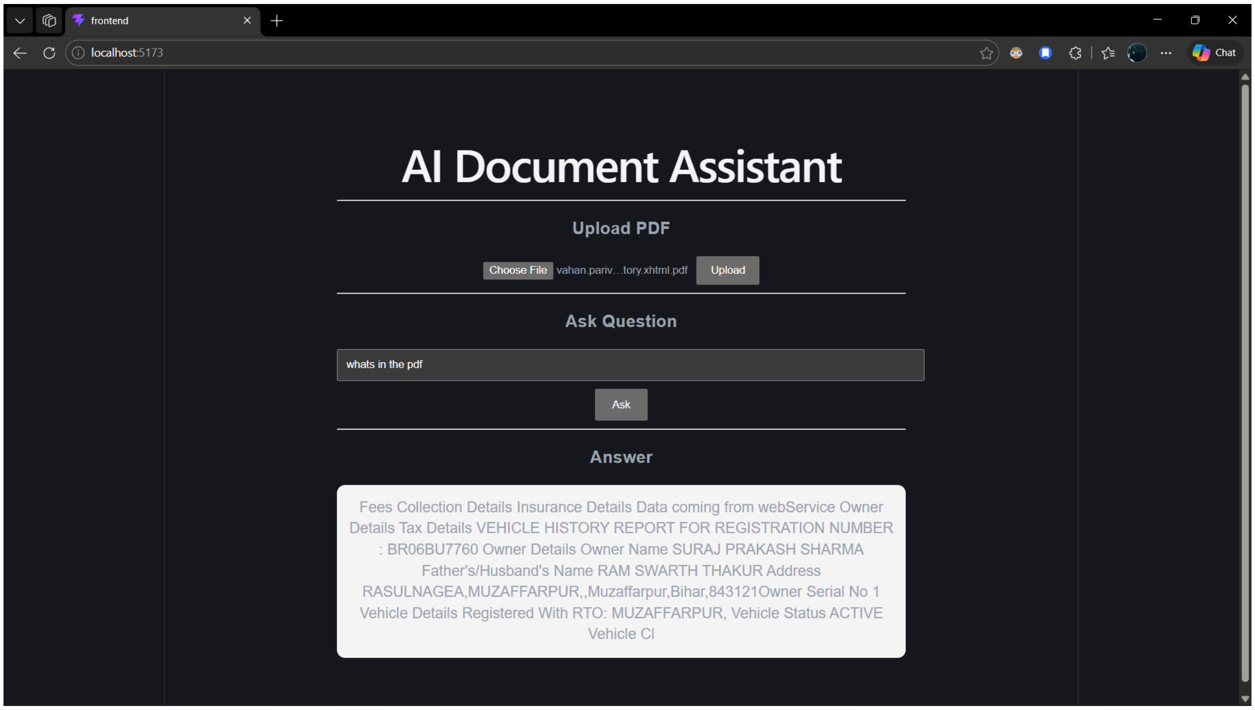
Frontend (React) → Spring Boot Backend → PDF Extraction → Chunking → MySQL Database → Retrieval-Based Response Generation

3. Application Screenshots

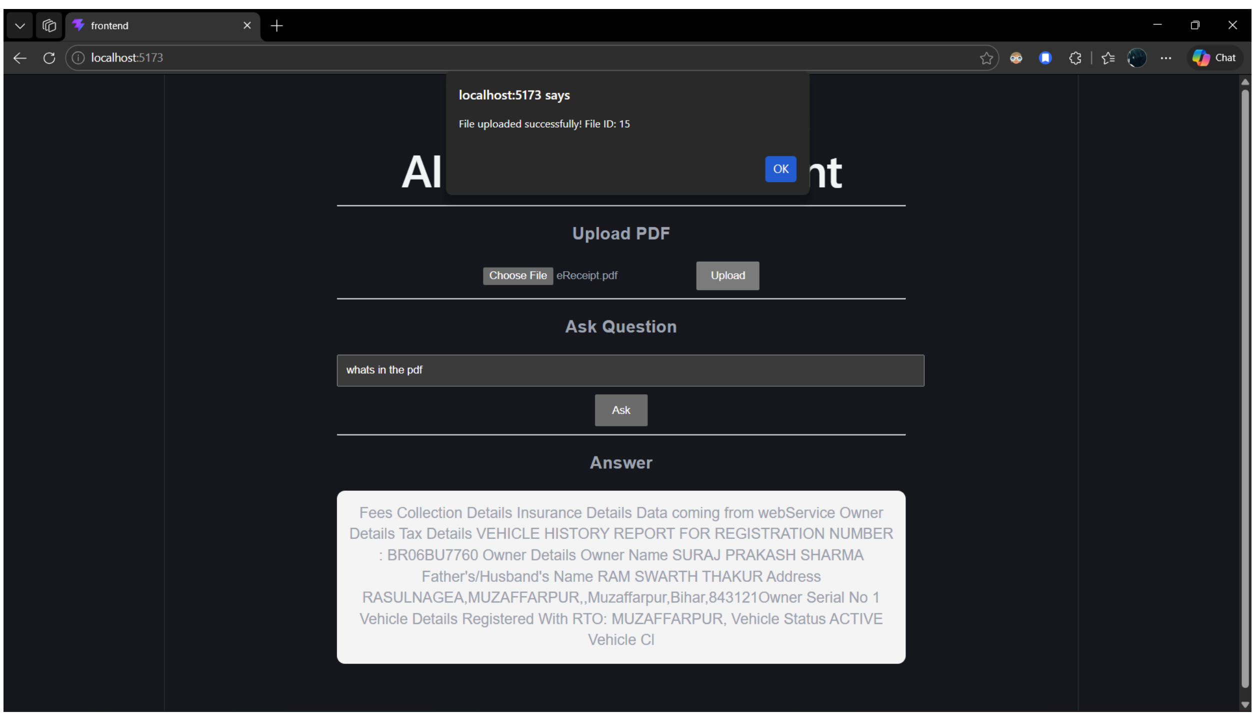
Main Application UI



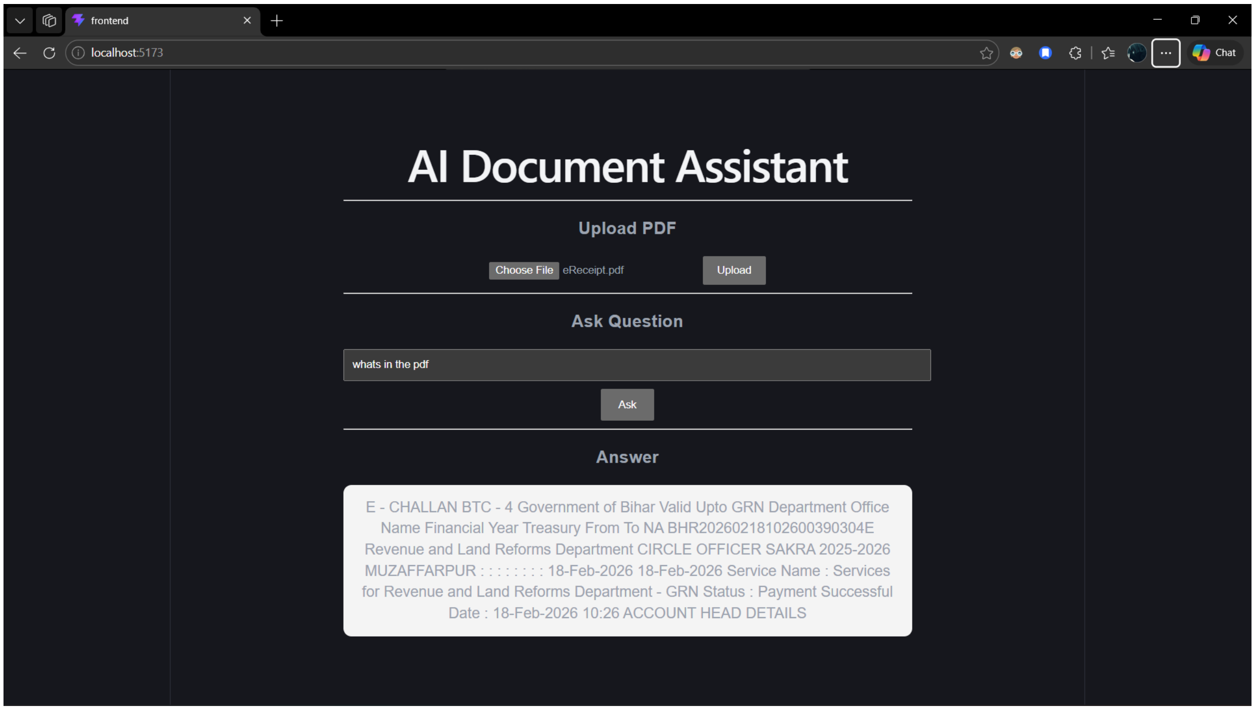
Question Answering Example



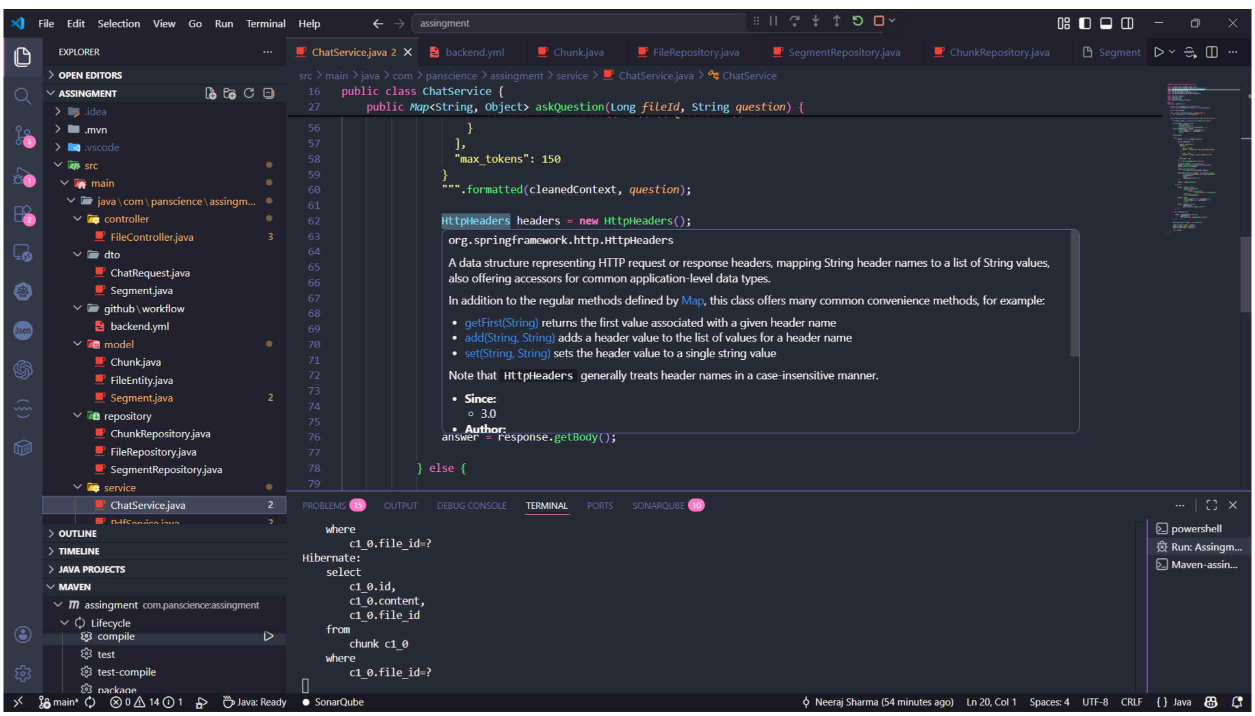
File Upload Success



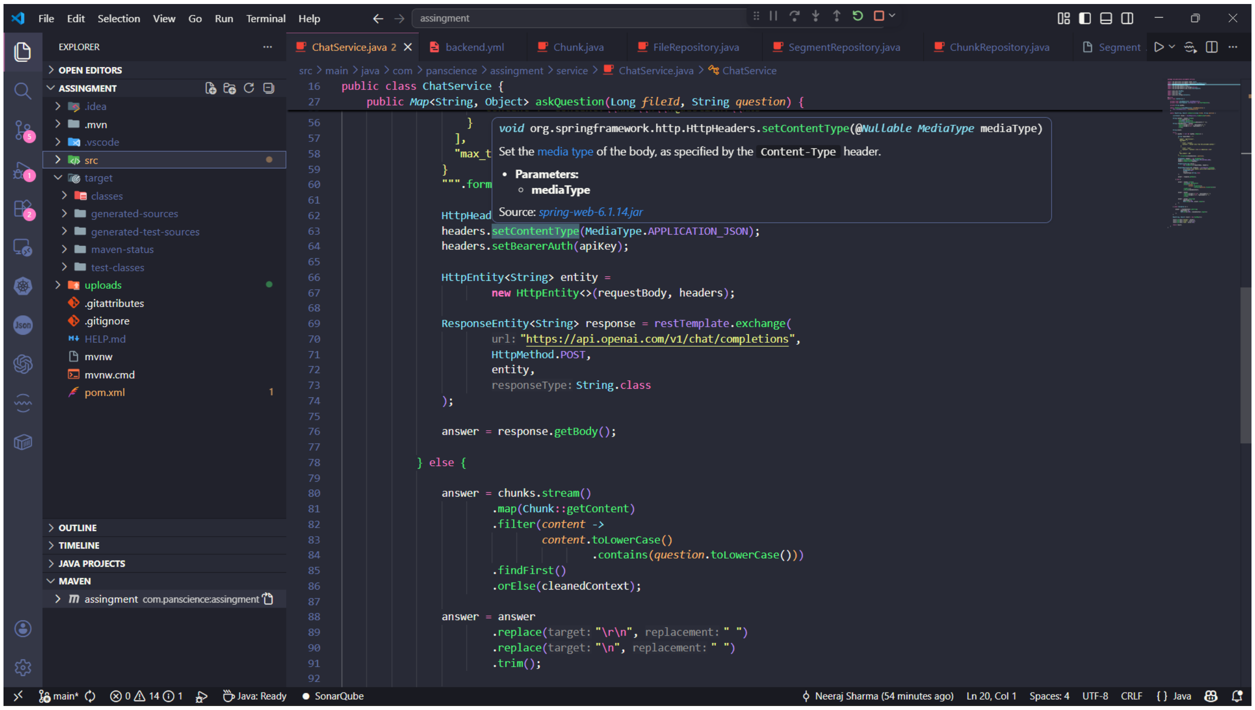
Processed Answer Response



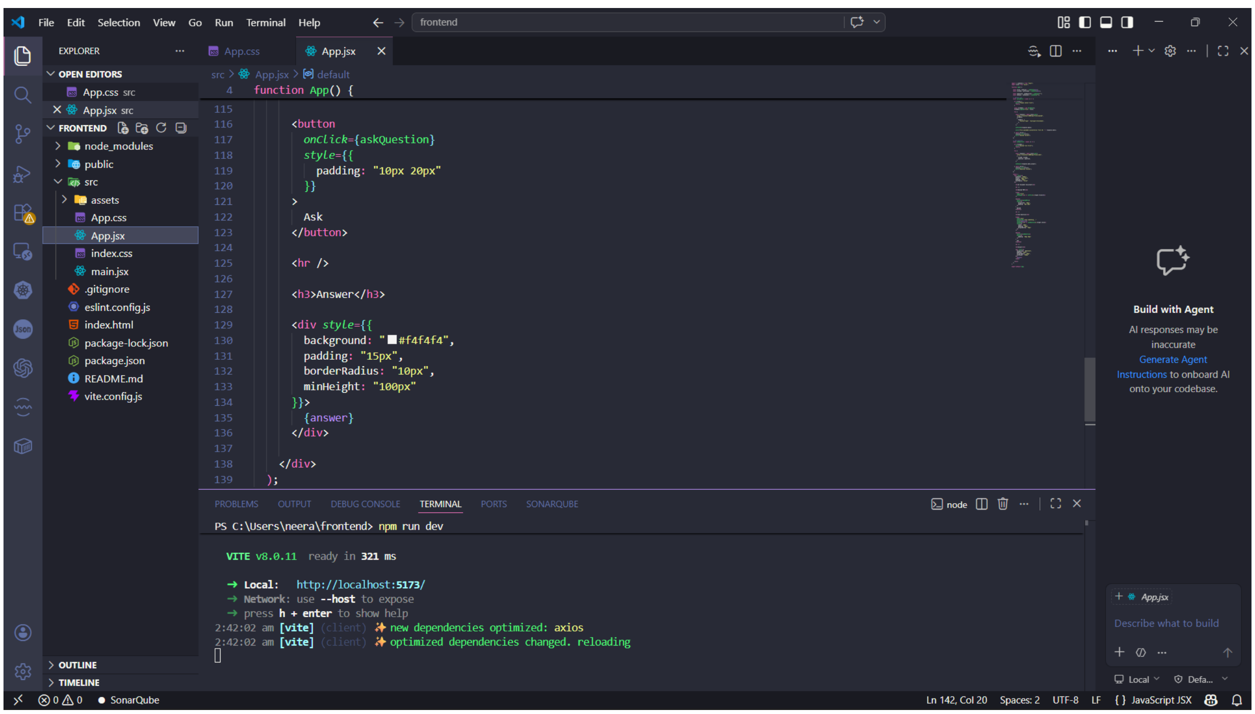
Backend Service Implementation



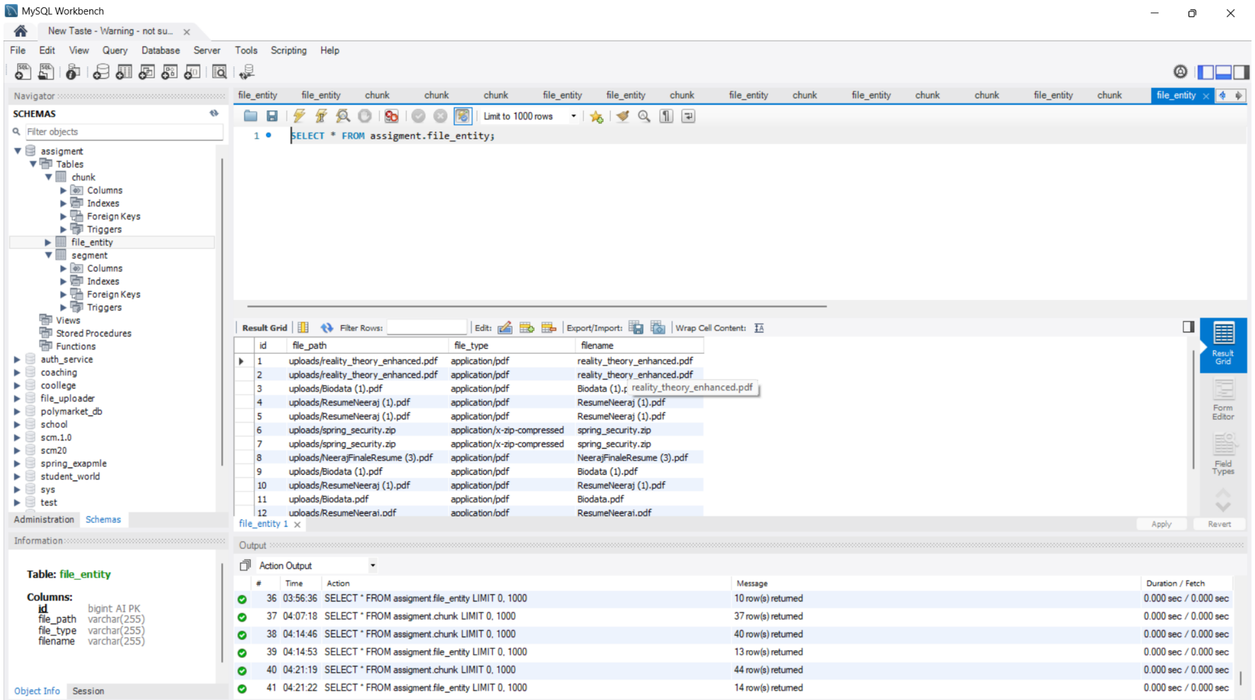
Frontend React Implementation



MySQL File Metadata Table



MySQL Chunk Storage Table



MySQL Workbench
New Taste - Warning - not su...
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
assigment
Tables
chunk
Columns
Indexes
Foreign Keys
Triggers
file_entity
segment
Columns
Indexes
Foreign Keys
Triggers
Views
Stored Procedures
Functions
auth_service
coaching
coollege
file_uploader
polymarket_db
school
scm.1.0
scm20
spring_exapmle
student_world
sys
test
Administration Schemas
Information
Table: file_entity
Columns:
id bigint AI PK
file_path varchar(255)
file_type varchar(255)
filename varchar(255)
Object Info Session
SELECT * FROM assigment.file_entity;
Limit to 1000 rows
Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:
id file_path file_type filename
1 uploads/reality_theory_enhanced.pdf application/pdf reality_theory_enhanced.pdf
2 uploads/reality_theory_enhanced.pdf application/pdf reality_theory_enhanced.pdf
3 uploads/Biodata (1).pdf application/pdf Biodata (1).pdf
4 uploads/ResumeNeeraj (1).pdf application/pdf ResumeNeeraj (1).pdf
5 uploads/ResumeNeeraj (1).pdf application/pdf ResumeNeeraj (1).pdf
6 uploads/spring_security.zip application/x-zip-compressed spring_security.zip
7 uploads/spring_security.zip application/x-zip-compressed spring_security.zip
8 uploads/NeerajFinaleResume (3).pdf application/pdf NeerajFinaleResume (3).pdf
9 uploads/Biodata (1).pdf application/pdf Biodata (1).pdf
10 uploads/ResumeNeeraj (1).pdf application/pdf ResumeNeeraj (1).pdf
11 uploads/Biodata.pdf application/pdf Biodata.pdf
12 uploads/ResumeNeeraj.pdf application/pdf ResumeNeeraj.pdf
file_entity 1
Apply Revert
Output
Action Output
Time Action Message Duration / Fetch
36 03:56:36 SELECT * FROM assigment.file_entity LIMIT 0, 1000 10 row(s) returned 0.000 sec / 0.000 sec
37 04:07:18 SELECT * FROM assigment.chunk LIMIT 0, 1000 37 row(s) returned 0.000 sec / 0.000 sec
38 04:14:46 SELECT * FROM assigment.chunk LIMIT 0, 1000 40 row(s) returned 0.000 sec / 0.000 sec
39 04:14:53 SELECT * FROM assigment.file_entity LIMIT 0, 1000 13 row(s) returned 0.000 sec / 0.000 sec
40 04:21:19 SELECT * FROM assigment.chunk LIMIT 0, 1000 44 row(s) returned 0.000 sec / 0.000 sec
41 04:21:22 SELECT * FROM assigment.file_entity LIMIT 0, 1000 14 row(s) returned 0.000 sec / 0.000 sec

4. API Endpoints

POST /api/files/upload
Uploads PDF documents to the backend server.

POST /api/files/chat
Accepts file ID and question, then returns context-based answer.

5. Challenges Faced

• Handling PDF text extraction reliably
• Managing chunk-based retrieval logic
• Integrating React frontend with Spring Boot backend
• Configuring Maven, Lombok, and MySQL dependencies
• Implementing CI/CD workflow using GitHub Actions

6. Future Improvements

• Semantic vector search integration
• Authentication and authorization
• Audio/video document support
• Advanced AI model integration
• Cloud deployment support

7. Conclusion

This project demonstrates a complete full-stack AI document assistant system using modern backend, frontend, database, and DevOps technologies. The application successfully processes PDF documents, stores extracted content, and provides contextual responses based on uploaded files.